

```
[Pipeline] stage
[Pipeline] { (init_and_plan)
[Pipeline] sh
[workspace] Running shell script
+ sudo terraform init /opt/bitnami/apps/jenkins/jenkins_home/sand-terraform-dep
[0m][1mInitializing modules...[0m
- module.asg
  Getting source "../terraform-modules/aws_asg"
- module.asg_lifecycle_hook_scale_in
  Getting source "../terraform-modules/aws_asg_lifecycle_hook"
- module.asg_lifecycle_hook_scale_out
  Getting source "../terraform-modules/aws_asg_lifecycle_hook"
- module.asg_policy
  Getting source "../terraform-modules/aws_asg_policy"
- module.cw
  Getting source "../terraform-modules/aws_cw"

[0m][1mInitializing the backend...[0m
[0m][32m
Successfully configured the backend "consul"! Terraform will automatically
use this backend unless the backend configuration changes.[0m

[0m][1mInitializing provider plugins...[0m
- Checking for available provider plugins on https://releases.hashicorp.com...
- Downloading plugin for provider "aws" (1.43.0)...

The following providers do not have any version constraints in configuration,
so the latest version was installed.

To prevent automatic upgrades to new major versions that may contain breaking
changes, it is recommended to add version = "... constraints to the
corresponding provider blocks in configuration, with the constraint strings
suggested below.

* provider.aws: version = "~> 1.43"
```

Figure 9: Jenkins log

triggering a continuous integration job.

4. If you have an existing Jenkins server, ensure that the pipeline plugin is installed on it. Otherwise, go to 'Manage plugins' and install the pipeline plugin.
5. In this project, Consul is being used as a remote back-end for state storing and state locking. It is not recommended to use local state when multiple people are involved in the project and for production deployments. It is good to use a remote back-end that provides highly available storage with state lock functionalities to avoid having the state being written by multiple users at the same time.
6. If you do not have a Consul key-value store in your environment, just pull a Consul Docker image and set up a single node cluster. If it is a production deployment, set up a distributed key-value store.
7. Create an application in Slack and note down the Slack integration details for configuring it in a Jenkins file.
8. Configure your provider and backend details in the main Terraform configuration file either by an environment variable or persisting in a repo. In my case, I am going to provision a resource in AWS and my CI server is hosted in AWS. So I am assigning an IAM role to my server with sufficient privileges.
9. Create a new project in Jenkins by using the pipeline plugin.
10. Add the Jenkins file where the pipeline stages are defined. Save the job and trigger it manually for testing. Then apply changes to the configuration, commit the changes to BitBucket and ensure the job is automatically triggered. Check the Jenkins log for more details about the job.

```
//Jenkins File
import groovy.json.JsonOutput
```

fetch_latest_code	install_deps	init_and_plan	approve	apply_changes	Declarative: Post Actions
7s	3s	656ms	236ms	25ms	43ms
7s	2s	559ms failed	25ms failed	24ms failed	41ms
7s	2s	567ms failed	26ms failed	25ms failed	45ms
7s	2s	560ms failed	24ms failed	27ms failed	40ms
6s	2s	568ms failed	24ms failed	24ms failed	39ms
6s	308ms failed	24ms failed	23ms failed	24ms failed	42ms
6s	2s	559ms failed	28ms failed	25ms failed	38ms
8s	4s	582ms failed	28ms failed	27ms failed	55ms
11s	4s	571ms failed	25ms failed	25ms failed	44ms
8s	3s	1s	1s (paused for 1d 0h) failed	25ms failed	42ms
8s	3s	1s	603ms (paused for 1d 1h) failed	28ms failed	47ms
7s	3s	1s	(paused for 1d 1h) 30s	25ms	41ms

Figure 10: Build history

```
//git env vars
env.git_url = 'https://user@bitbucket.org/user/terraform-ci.git'
env.git_branch = 'master'
env.credentials_id = '1'
```

```
//slack env vars
env.slack_url = 'https://hooks.slack.com/services/SDKJSDKS/SDSDJSDK/SDKJSDKDS23434SDSDLCMLC'
env.notification_channel = 'my-slack-channel'
```

```
//jenkins env vars
env.jenkins_server_url = 'https://52.79.46.98'
env.jenkins_node_custom_workspace_path = "/opt/bitnami/apps/jenkins/jenkins_home/${JOB_NAME}/workspace"
env.jenkins_node_label = 'master'
env.terraform_version = '0.11.10'
```

```
def notifySlack(text, channel, attachments) {
    def payload = JsonOutput.toJson([text: text,
        channel: channel,
        username: "Jenkins",
        attachments: attachments
    ])
    sh "export PATH=/opt/bitnami/common/bin:$PATH && curl -X
```